

Retrieval–Reasoning Processes for Multi-hop Question Answering: A Four-Axis Design Framework and Empirical Trends

Yuelyu Ji¹, Zhuochun Li¹, Rui Meng², Daqing He¹

¹School of Computing and Information, University of Pittsburgh

²Google Cloud AI Research, Sunnyvale, CA, USA

Abstract

Multi-hop question answering (QA) requires a system to *retrieve* and *reason* over multiple pieces of evidence. Recent advances in retrieval-augmented generation (RAG) and agent-style systems have achieved strong performance on benchmarks such as HotpotQA, 2WikiMultiHopQA, and MuSiQue. However, the *retrieval–reasoning process*, which explicitly interleaves evidence acquisition and inference decisions, is often left implicit and treated as an implementation detail. We propose a design framework that treats this process as the unit of analysis and describes it along four axes: the overall execution plan, the index structure, the next-step execution plan, and the stop/continue criteria. Using this framework, we synthesize reported empirical results from representative multi-hop QA systems on standard datasets and analyze how different procedural design choices are discussed in relation to effectiveness, efficiency, and explanation faithfulness across the literature. We then distill a set of open problems for future retrieval–reasoning agents, including coupling retrieval with index structures, learning generalizable control policies, and designing robust stopping signals.

1 Introduction

Multi-hop question answering (QA) has become a core testbed for machine reasoning. Compared to single-hop QA, where a single passage often suffices, multi-hop benchmarks such as HotpotQA (Yang et al., 2018), 2WikiMultiHopQA (Ho et al., 2020), and MuSiQue (Trivedi et al., 2022) require systems to combine several pieces of evidence, compare attributes across entities, or follow chains of relations that span multiple documents (Shi et al., 2024). Many of these datasets annotate supporting sentences and sometimes explicit reasoning paths, making them natural grounds for analyzing how systems actually reason.

Large language models (LLMs) equipped with retrieval have become the dominant approach for such tasks. Retrieval-augmented generation (RAG) couples a retriever with an LLM and has achieved strong performance on knowledge-intensive QA (Lewis et al., 2020; Gao et al., 2023b), while more recent agent-style systems model question answering as a sequence of actions over tools and documents (He et al., 2024; Kim et al., 2024). However, most of this literature focuses on architectural or training innovations and treats *how retrieval and reasoning interact* as an implementation detail. Concretely, the sequence of retrieval calls, intermediate hypotheses, and stopping decisions is rarely made explicit or analyzed systematically. This omission matters because many multi-hop benchmarks are brittle to single-pass retrieval under distractors or long contexts, motivating explicit retrieval–reasoning loops that adapt queries, evidence selection, and stopping over time.

At the algorithmic level, what distinguishes multi-hop QA from single-hop QA is not only the presence of retrieval, but how retrieval and reasoning are coupled over time. These procedural decisions directly shape reported trade-offs in accuracy, efficiency, and robustness, and motivate treating the retrieval–reasoning process as an explicit object of analysis.

Unlike existing surveys that primarily organize methods by modules, stages, or training loops, our framework treats the execution procedure as the unit of analysis. This emphasizes how retrieval, reasoning, control, and stopping decisions are interleaved over time, rather than discussed as separable components. As a result, design choices that are often presented as method-specific heuristics—e.g., controller triggers and stop/continue signals—become alignable across model families under a shared axis-based schema.

Our goal is to make these procedural choices explicit and comparable across systems rather than

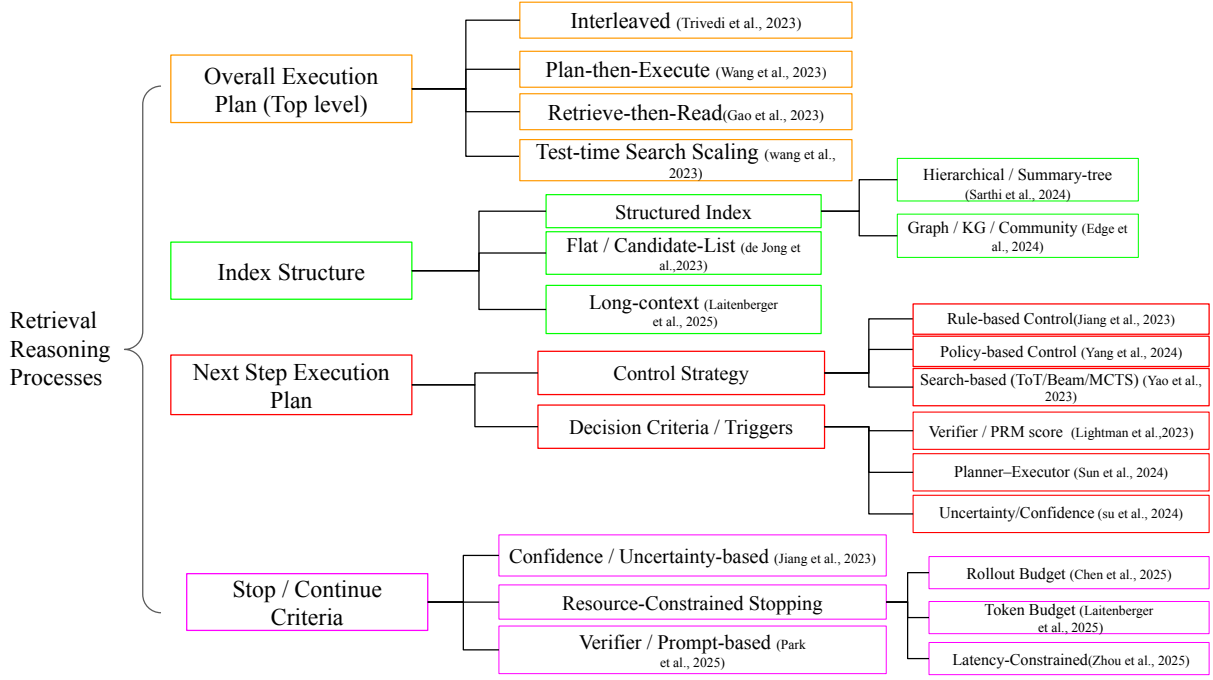


Figure 1: Overview of the four design axes we use to describe retrieval–reasoning processes in multi-hop QA: (A) overall execution plan, (B) index structure, (C) next-step execution plan, and (D) stop/continue criteria. The framework abstracts over concrete model architectures and datasets.

to claim causal superiority. We organize the survey around four questions: (RQ1) why make the procedure explicit, (RQ2) what are the key procedural design axes, (RQ3) what trade-offs are reported, and (RQ4) what open challenges remain.

2 Background

2.1 Multi-hop QA Datasets

Multi-hop question answering (QA) requires a system to combine information from multiple pieces of evidence to answer a single question. Datasets such as HotpotQA, 2WikiMultiHopQA, and MuSiQue are explicitly designed to test this ability (Yang et al., 2018; Ho et al., 2020; Trivedi et al., 2022). HotpotQA provides Wikipedia-based questions with sentence-level supporting facts and adversarial distractor paragraphs (Yang et al., 2018). 2WikiMultiHopQA mixes structured and unstructured evidence and includes annotated reasoning paths (Ho et al., 2020), while MuSiQue composes single-hop questions into 2–4 hop questions and controls for artifacts and disconnected reasoning (Trivedi et al., 2022). More recent datasets, such as MEQA and FanOutQA, further emphasize multi-document aggregation and the completeness and consistency of supporting evidence (Li et al., 2024a; Zhu et al., 2024).

2.2 Existing Surveys and Remaining Gaps

Several surveys and tutorials organize the rapidly growing literature on multi-hop QA and reasoning-capable LLMs. Mavi et al. (2024) review multi-hop QA datasets and model architectures, focusing on how graphs, question decomposition, and memory mechanisms support complex reasoning. Other surveys discuss reasoning LLMs more broadly, covering prompting strategies, training recipes, and cognitive perspectives (Li et al., 2025b; Sui et al., 2025), or analyze when parametric knowledge needs to be complemented by external tools and retrieval (Li et al., 2025a). These overviews cover many *ingredients* relevant to our focus (decomposition, retrieval modules, memory, verification), but they typically organize the space around *components*, stages, or model families (e.g., graph-based vs. decomposition-based; RAG vs. non-RAG) rather than treating the *execution procedure*—the step-by-step coupling between retrieval and reasoning—as the primary unit of analysis (Gao et al., 2023c).

Consequently, procedural questions such as *when retrieval is invoked relative to reasoning*, *how evidence is organized and retained under context budgets*, *how the next action is chosen*, and *how stop/continue decisions are made* often appear

only implicitly as method-specific design details or scattered heuristics. In particular, stop/continue is rarely separated as a first-class design choice, and the literature is not mapped into an execution-oriented coding schema enabling aligned comparisons across model families. In contrast, our framework takes the execution procedure itself as the unit of analysis and makes retrieval, reasoning, control, and stopping decisions explicitly comparable along shared axes.

3 Why an Explicit Retrieval–Reasoning Process?

Our first research question (**RQ1**) asks why it is useful to treat multi-hop QA as an explicit retrieval–reasoning *procedure* rather than as a one-time retrieve-and-read. The core reason is that, for knowledge-intensive multi-step questions, the *procedure*—when to retrieve, how to form the next query, what evidence to keep under budgets, and when to stop—is often the main lever that changes both accuracy and failure modes. For example, IRCot improves multi-step QA by *interleaving* retrieval with chain-of-thought, i.e., by changing the retrieval–reasoning interaction rather than only scaling the reader (Trivedi et al., 2023). Moreover, agentic QA methods such as ReAct make tool calls and evidence usage explicit, turning the trajectory itself into a process-level account of how an answer is obtained (Yao et al., 2022). Finally, recent work treats stopping as a first-class decision (instead of a fixed hop budget), highlighting that *continue* vs. *stop* is part of the procedure and directly controls cost and robustness (Park et al., 2025). Our coding of 252 recent multi-hop QA/RAG papers (Appendix B) further suggests that many recent advances can be reinterpreted as innovations in retrieval–reasoning coupling: 104 of the 252 papers (41%) are rated high-fit on this dimension, motivating a survey organized around procedural design choices and their trade-offs.

4 Design Axes of Retrieval–Reasoning Procedures

To answer **RQ2**, we describe multi-hop QA systems in terms of the four design axes shown in Figure 1. Each axis abstracts over many concrete modules and implementations; our goal is to provide a shared vocabulary for characterizing execution procedures.

4.1 Overall Execution Plan

The overall execution plan specifies *when* retrieval is invoked relative to reasoning. We highlight four recurring patterns that correspond to the subpanels in Figure 2: retrieve-then-read, interleaved retrieval, plan-then-execute, and test-time search scaling.

Retrieve-then-Read. Classical retrieval-augmented QA pipelines retrieve a fixed set of passages given the original question, then feed them once into a reader or generator. Examples include DrQA-style open-domain QA (Chen et al., 2017), DPR + BERT readers (Karpukhin et al., 2020), FiD-style fusion-in-decoder models (Izacard and Grave, 2021), and the original RAG model (Lewis et al., 2020). Multi-hop extensions such as MDR (Qi et al., 2019) follow the same pattern: the retriever is called a small fixed number of times, and the reader operates in a single, long-context pass.

Interleaved Retrieval and Reasoning. Interleaved plans alternate between reasoning steps and retrieval calls. Self-Ask (Press et al., 2023) and ReAct (Yao et al., 2022) are early examples that prompt the LLM to produce intermediate thoughts and explicit SEARCH/LOOKUP actions. IRCot (Trivedi et al., 2023) couples chain-of-thought with retrieval on HotpotQA and MuSiQue. FLARE (Jiang et al., 2023) and DRAGIN (Su et al., 2024) further refine when and how new queries are issued during generation. Recent analyses of retrieval failure modes in such loops propose targeted query reformulation and filtering for multi-hop RAG.

Plan-then-Execute. Plan-then-execute methods first produce an explicit reasoning or decomposition plan, and only then execute retrieval and answering steps according to that plan. Decomposed Prompting (Khot et al., 2022), Plan-and-Solve (Wang et al., 2023), and Least-to-Most Prompting (Zhou et al., 2022) decompose questions into sub-questions that can be answered sequentially. QDMR (Wolfson et al., 2020) and recent QDMR-based multi-hop QA systems use structured decomposition graphs as plans. In retrieval settings, PAR-RAG (Zhang et al., 2025) and BEAM Retrieval (Zhang et al., 2024) first plan candidate evidence chains and then execute retrieval over them, sometimes with verifier feedback.

Test-Time Search Scaling. A more aggressive family of overall execution plans treats the whole retrieval–reasoning process as a search problem and scales performance by exploring many can-

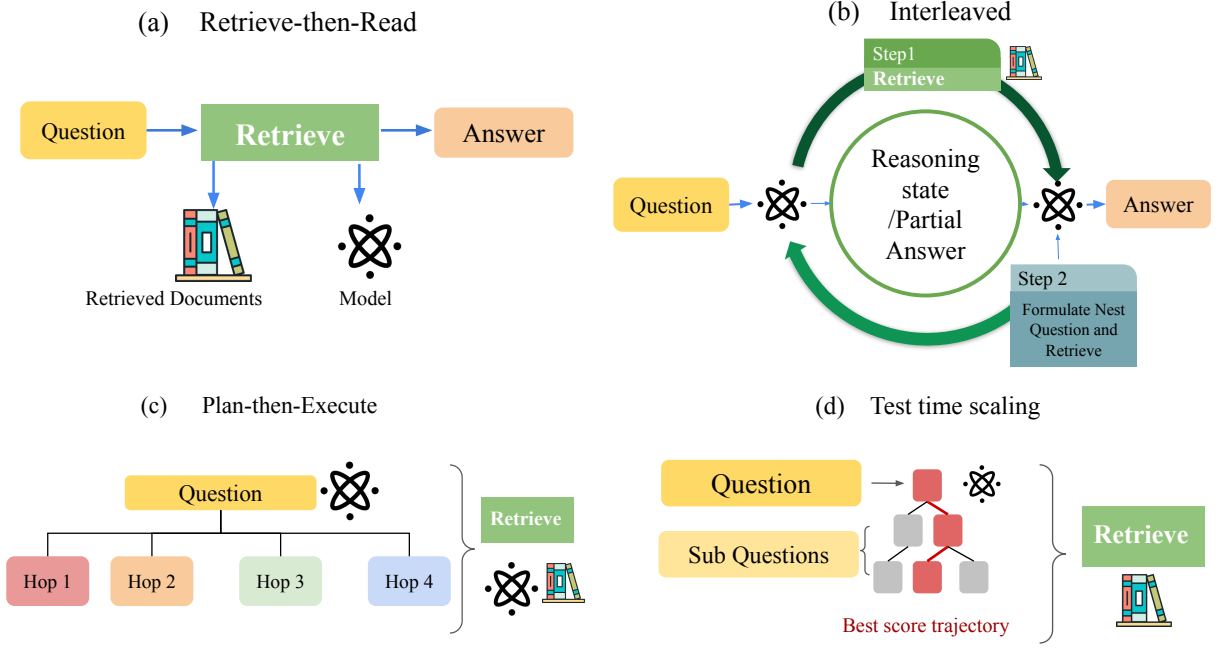


Figure 2: Overall execution plans for retrieval-reasoning in multi-hop QA (Axis A of our framework). (a) *Retrieve-then-Read*: a single retrieval pass feeds the reader model. (b) *Interleaved*: the model alternates reasoning with additional retrieval calls, updating a partial answer/state. (c) *Plan-then-Execute*: the model first decomposes the question into sub-hops, then retrieves and answers for each hop following the plan. (d) *Test-time search scaling*: the model explores multiple candidate reasoning trajectories over the question/sub-questions and selects the best-scoring path for retrieval and answering.

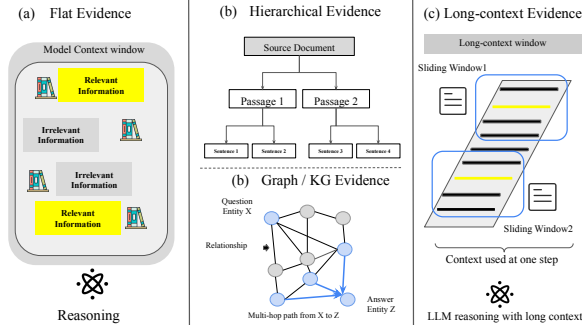


Figure 3: Index structures for retrieval-reasoning (Axis B): flat passage lists, hierarchical / tree-structured indices, graph-based indices, and long-context storage.

candidate trajectories at test time. Tree-of-Thoughts (Yao et al., 2023) and Graph-of-Thoughts (Besta et al., 2024) use explicit search over thought trees and graphs. MindStar (Kang et al., 2024), MCTS-DPO (Xie et al., 2024), and Monte-Carlo-tree self-refinement methods apply MCTS-style search guided by value functions or process reward models such as ReARTeR (Sun et al., 2025). In knowledge-intensive settings, RAP (Hao et al., 2023) plans over world models, and several GraphRAG (Han et al., 2024) variants explore multiple graph traversal paths before committing to an answer.

4.2 Index Structure

Index structure specifies *how* external knowledge is organized and accessed. Figure 3 distinguishes four common forms: flat candidate lists, hierarchical or summary trees, graph- or KG-based structures, and long-context evidence units.

Flat / Candidate-List Indices. The default in most multi-hop QA systems is a flat collection of passages or documents indexed by BM25 or dense embeddings. DrQA (Chen et al., 2017), DPR (Karpukhin et al., 2020), FiD (Izacard and Grave, 2021), and RAG adopt this structure in open-domain QA. Multi-hop retrievers such as MDR (Xiong et al., 2020), HopRetriever (Li et al., 2021), and step-wise dense retrievers for HotpotQA and 2WikiMultiHopQA operate over flat indices, even when they train separate retrievers per hop.

Hierarchical / Summary-Tree Indices. Hierarchical indices organize documents into multi-level clusters or summary trees, allowing systems to retrieve coarse summaries first and zoom in on relevant subtrees. RAPTOR (Sarathi et al., 2024) builds recursive summary trees and shows that retrieving via tree nodes can substantially reduce tokens while preserving answer accuracy. Other long-document QA systems build section- or chapter-based hierar-

System	A	B	C	D	Benchmarks (typical)
DrQA / DPR+FiD	R	F	H	B	NQ, TriviaQA (Chen et al., 2017; Karpukhin et al., 2020; Izacard and Grave, 2021)
IRCoT	I	F	H	B	HotpotQA, MuSiQue, IIRC (Trivedi et al., 2023, 2022)
DEC	I	F	P	B/V	HotpotQA (Ji et al., 2025)
Beam Retrieval	P	F	P	B	HotpotQA, 2Wiki, MuSiQue (Zhang et al., 2024)
PAR-RAG	P	F/G	V	B	HotpotQA, 2Wiki (Zhang et al., 2025)
RAPTOR	R/I	H	H	B	Long-doc QA, Hotpot variants (Sarthi et al., 2024)
LongRAG	R/I	L	H	B	HotpotQA, ODQA (Jiang et al., 2024; Zhao et al., 2024)
GraphRAG	I	G	H/P	B	Sensemaking, MHQA (Edge et al., 2024)
KG-o1	P/I	G	P/V	B	2Wiki, KGQA (Wang et al., 2025a)
SIM-RAG	I	F	P/V	C/L	Multi-round RAG (Xu et al., 2025)
Stop-RAG	I/P	F	P	L	Iterative RAG / MHQA (Park et al., 2025)

Table 1: Representative systems mapped into our four-axis framework. A: overall execution plan (R=retrieve-then-read, I=interleaved, P=plan-then-execute, S=test-time search scaling). B: index structure (F=flat, H=hierarchical, G=graph/KG, L=long-context). C: control (H=hand-written rules, P=learned policy, V=verifier-gated, S=search-based). D: stopping (B=budget, C=confidence-based, V=verifier-gated, L=learned/value-based).

chies and retrieve using both coarse and fine nodes (Xia et al., 2025). LongRAG (Zhao et al., 2024) uses long retrieval units (e.g., 4K tokens) as coarse segments and lets the LLM decide which parts to attend to, effectively creating a two-level index structure. Hierarchical community-based indices in GraphRAG also fall into this category when communities are organized in a tree (Edge et al., 2024).

Graph / KG / Community Structures. Graph-based structures connect entities, documents, or passages via explicit edges, enabling path-based reasoning and community-level summarization. Early multi-hop QA work over KGs includes GraftNet (Sun et al., 2018) and PullNet (Sun et al., 2019). More recent GraphRAG systems construct entity and document graphs from text and perform retrieval along graph edges or communities (Edge et al., 2024). KG-augmented RAG approaches such as KG-FiD (Yu et al., 2022), UniKGQA (Jiang et al., 2022), and KG-o1 (Wang et al., 2025a) extend this idea by combining KG traversal with text retrieval for multi-hop QA.

Long-Context Evidence. Instead of relying on a structured index, some methods rely on long-context LLMs and store very long segments or entire documents as retrieval units. LongRAG explicitly proposes long units to reduce the number of

hops needed to cover the required evidence (Zhao et al., 2024; Lee et al., 2025; Das et al., 2025). LongT5 and other long-context models similarly retrieve and process long sequences in a retrieve-and-read style (Guo et al., 2022; Li et al., 2024b).

4.3 Next-Step Execution Plan

The *next-step execution plan* specifies how a system chooses its next action—retrieve again, expand candidates, verify, backtrack, or answer—given the current state. We separate this component into (1) a control strategy, which proposes candidate actions, and (2) decision criteria / triggers, which determine whether to continue expanding or to revise/hand off control (termination is covered in Axis D).

Rule-based control. Many systems hard-code a reasoning control pattern: a fixed alternation of REASON and SEARCH steps, simple heuristics for query reformulation, and manually tuned filters over retrieved passages. Self-Ask, ReAct, IRCoT, and DEC all fall in this family, using prompts and hand-designed rules to decide when to ask a follow-up question, when to call the retriever, and when to output an answer (Press et al., 2023; Yao et al., 2022; Trivedi et al., 2023; Ji et al., 2025). Rule-based controllers are easy to implement and robust in small-data regimes, but they do not learn from failures and often need re-tuning when the domain or retriever changes.

Policy-based control. Here the next action is chosen by a learned policy that observes the question, retrieved evidence, and partial reasoning trace. Examples include beam-style multi-hop retrievers with learned expansion policies (Zhang et al., 2024), RAG agents trained to select between tools or retrieval templates via supervision or reinforcement learning (Yang et al., 2024; Park et al., 2025), and multi-round RAG systems whose controllers are fine-tuned on inner-monologue trajectories (Xu et al., 2025).

Search-based control. A third pattern treats next-step selection as search over partial trajectories. Tree-of-Thoughts explores a tree of reasoning states and uses value- or vote-based scores to decide which branches to expand (Yao et al., 2023). MindStar and related MCTS-style methods perform Monte-Carlo search over tool-use and reasoning sequences, guided by process reward models or learned value functions (Kang et al., 2024; Xie et al., 2024). Beam-style search over plans or KG paths similarly expands multiple candidate chains in parallel before committing to an answer (Zhang

et al., 2024; Jiang et al., 2022). These approaches can substantially improve long-horizon success, but incur higher inference cost and rely on strong scoring signals.

Verifier / PRM-score triggers. Verifier-gated loops insert an explicit judge—an NLI-style checker, a process reward model, or a self-evaluation module—and use its score as a trigger. Process reward models that score intermediate reasoning steps (Lightman et al., 2023) and hallucination detectors such as SelfCheckGPT (Manakul et al., 2023) exemplify this pattern: low-scoring answers or chains trigger extra retrieval or revision instead of termination. Tool-centric systems such as TaskMatrix-style agents also rely on verifier-like scoring over tool outputs to decide whether a subgoal is solved or whether another tool call is needed (Liang et al., 2024). In RAG settings, verifier / PRM triggers are often combined with sufficiency critics, as in SIM-RAG and related multi-round controllers (Xu et al., 2025; Zhang et al., 2025).

Planner-executor triggers. Some agents separate planning and execution and use the plan’s status as an explicit trigger. Planner-executor architectures first generate a structured plan (sub-questions, tool calls, or graph operations) and then execute it step-wise, revising or extending the plan when execution fails or when critical preconditions are not met (Khot et al., 2022; Wang et al., 2025b; Sun et al., 2025). In this view, the decision criterion is whether the current plan still appears feasible and sufficiently specified; negative feedback triggers re-planning or refinement, while successful plan completion triggers termination. This pattern is especially common in web and tool agents that must coordinate several API calls (He et al., 2024).

Uncertainty / confidence triggers. Finally, uncertainty-aware triggers estimate how likely the current answer (or intermediate state) is to be correct or sufficiently supported, and continue retrieving or reasoning only while confidence is below a threshold. Signals include calibrated answer probabilities, disagreement across sampled chains, and explicit “uncertainty of thoughts” scores (Hu et al., 2024; Sui et al., 2025). Dynamic context-cutoff policies similarly stop reading long inputs once new tokens are unlikely to change the prediction (Xie et al., 2025). In multi-round RAG, such confidence signals are often combined with learned sufficiency critics or value-based controllers (Xu et al., 2025; Park et al., 2025), yielding adaptive-depth retrieval loops that trade a small amount of

computation for more stable behavior under distribution shift.

4.4 Stop / Continue Criteria

Stop / continue criteria specify *when* the retrieval-reasoning procedure terminates. In multi-hop QA, stopping too early misses necessary evidence, while stopping too late inflates prompt length and latency and increases exposure to distractors.

Resource-constrained stopping. Many systems ignore the detailed state and instead impose hard budgets on search. A first subtype is a *rollout budget*: interleaved agents such as Self-Ask, ReAct, IRCot, and DRAGIN restrict the number of reasoning or tool-use steps (e.g., at most K SEARCH calls or H decomposition steps) and must answer once the budget is exhausted (Press et al., 2023; Yao et al., 2022; Trivedi et al., 2023; Su et al., 2024). A second subtype is a *token budget*, where multi-hop RAG pipelines (e.g., LongRAG, DEC, and budget-aware chain-of-thought methods) fix the maximum retrieved or prompt tokens per question and truncate extra evidence or thoughts when this limit is reached (Jiang et al., 2024; Ji et al., 2025; Han et al., 2025). A third subtype is *latency-constrained* stopping, where web or tool agents terminate once a wall-clock or API-call budget is consumed (Shen et al., 2024).

Confidence- / uncertainty-based stopping. A second family asks how *confident* the model (or a small auxiliary head) is in the current answer or intermediate state, and stops once uncertainty falls below a threshold. Representative signals include logit margins, entropy, and disagreement across self-consistent samples, as surveyed in work on “stop overthinking” and early-exit reasoning (Sui et al., 2025; Li et al., 2025b). Uncertainty-of-thought approaches explicitly estimate the uncertainty of candidate thoughts and decide whether to produce more intermediate steps or to answer directly (Hu et al., 2024), while dynamic context-cutoff methods learn when an LLM has read enough of a long input and can safely stop consuming further tokens (Xie et al., 2025).

Verifier- / prompt-based stopping. The third family inserts an explicit *checking* stage and delegates the stop / continue decision to a verifier. Rationale-refinement and fact-checking pipelines such as RARR, SelfCheckGPT, and Chain-of-Verification first generate a draft answer and then ask a verifier (often another LLM or an NLI-style model) whether each claim is sufficiently supported

by the retrieved evidence; a positive verdict triggers termination, while unsupported claims trigger additional retrieval, clarification questions, or answer revision (Gao et al., 2023a; Manakul et al., 2023; Dhuliawala et al., 2024). In multi-round RAG, sufficiency critics and value-based controllers such as SIM-RAG, Stop-RAG, DEC, and ReARTeR can be viewed as *learned verifiers*: they inspect the question, retrieved evidence, and inner-monologue trace and decide whether another retrieval round is expected to help, using QA accuracy or process rewards as supervision (Xu et al., 2025; Park et al., 2025; Ji et al., 2025; Sun et al., 2025).

Table 1 maps representative systems onto these four axes.

5 RQ3: Reported Empirical Tendencies in Retrieval–Reasoning Design

This section performs a pattern-mining synthesis of empirical findings as reported in the original papers. Because results are obtained under heterogeneous retrievers, LLMs, prompts, and budgets, we do not claim causal superiority of any axis choice. Instead, we summarize directional tendencies that (1) recur across multiple studies, or (2) are supported by within-paper ablations under matched budgets, and we treat cross-paper comparisons as suggestive rather than definitive.

Overall execution plan. Compared to retrieve-then-read (single-pass retrieval) baselines, several studies report an association between explicitly multi-step execution plans and higher answer F1 and joint answer–evidence scores, particularly in full-wiki and distractor-heavy settings (Trivedi et al., 2023). Interleaved plans that alternate chain-of-thought and retrieval, such as IRCOT-style interleaving, are reported to increase recall and joint F1 on HotpotQA, 2WikiMultiHopQA, and MuSiQue (Trivedi et al., 2023). Within the reported settings (often under matched budgets within the same study), plan–then–execute methods are reported to reach comparable gains while batching sub-queries (Zhang et al., 2024, 2025). Test-time search-scaling methods (e.g., Tree-of-Thoughts, MindStar, MCTS-style search) further boost long-horizon accuracy by exploring many candidate trajectories, but incur the highest latency and compute costs (Yao et al., 2023; Kang et al., 2024; Xie et al., 2024).

Index structure. Most multi-hop QA systems still operate over flat BM25 or dense passage col-

lections, and attribute gains mainly to changes in overall execution plan or control policy rather than radically different indices (Chen et al., 2017; Karpukhin et al., 2020; Trivedi et al., 2023; Zhang et al., 2024; Ji et al., 2025). Index structures are primarily discussed in relation to efficiency and path faithfulness. Hierarchical or summary-tree structures such as RAPTOR reduce context length on long-document QA by retrieving summaries and then drilling down to raw passages, typically preserving or slightly improving EM (Sarathi et al., 2024). Graph- and KG-based index structures (e.g., GraphRAG, KG-o1) organize entities and documents into graphs or communities and retrieve along paths, improving joint answer+path metrics and producing more coherent evidence chains, at the cost of graph construction and maintenance (Edge et al., 2024; Wang et al., 2025a).

From control and triggers (Axes C/D) to outcomes. For the next-step execution plan, our coded systems are consistent with three recurring patterns. First, moving from purely rule-based control with static budgets to learned or planner-style control tends to improve answer accuracy at similar average cost. Beam Retrieval and PAR-RAG replace hand-written rhythms with plan-driven controllers and beam-style expansion: Beam Retrieval reports 69.2 answer F1 / 91.4 support F1 on MuSiQue, and PAR-RAG reports 2Wiki and HotpotQA EM gains of roughly 15–30 points over their strongest non-planning baselines under matched or smaller budgets (Zhang et al., 2024, 2025). These results suggest that global, plan-aware controllers are especially beneficial when the evidence graph is rich and hop depth varies widely.

Second, combining policy-based control with verifier or PRM-style triggers is repeatedly reported to coincide with improved faithfulness and joint answer+evidence scores. Verifier-gated refinement in RARR and Chain-of-Verification reduces unsupported claims compared to single-pass RAG and vanilla CoT (Gao et al., 2023a; Dhuliawala et al., 2024). Similar trends appear in search-based controllers such as MindStar and MCTS-DPO, where PRM-guided tree search prunes low-scoring trajectories and improves math or strategy benchmarks at the cost of a small constant factor in inference time (Kang et al., 2024; Xie et al., 2024). In our framework, “policy/search + verifier/PRM” appears effective on noisy-retrieval or long-horizon tasks where hallucinated chains are common.

Third, uncertainty- or value-based stopping

(Axis D) helps controllers avoid both over- and under-search and stabilizes latency under shift (Xu et al., 2025; Park et al., 2025). Overall, the most robust designs in our corpus pair an adaptive controller (policy- or search-aware, Axis C) with verifier/PRM- or sufficiency-based triggers (Axis D), especially under noisy retrieval and long horizons. Static hop/token budgets can be competitive when tuned in-domain, but they transfer poorly across datasets with different hop and distractor profiles.

6 RQ4: Open Problems and Future Directions

Challenge 1: Aligning overall execution plans with index structures. Most systems fix an *index structure* (flat passages, hierarchical summaries, graphs, or long-context units) and then adopt an *overall execution plan* (retrieve-then-read, interleaved, plan-then-execute, or test-time search scaling) in an ad hoc manner. For example, IRCoT, Beam Retrieval, and DEC operate over flat indices and primarily innovate in scheduling and control (Trivedi et al., 2023; Zhang et al., 2024; Ji et al., 2025), whereas RAPTOR, LongRAG, and GraphRAG redesign the structure but often retain standard multi-hop schedules (Sarathi et al., 2024; Jiang et al., 2024; Edge et al., 2024). A principled mapping between structure and plan is still missing. Promising directions include factorial sweeps over execution plans and index structures under matched budgets, and controllers that choose among multiple (plan, structure) configurations based on query difficulty and resource constraints.

Challenge 2: Scalable and adaptive structured indices. Structured indices—such as summary trees (RAPTOR), long units (LongRAG), and graphs (GraphRAG, KG-o1)—can improve efficiency and faithfulness, but they are costly to build and maintain (Sarathi et al., 2024; Jiang et al., 2024; Edge et al., 2024; Wang et al., 2025a). Most work assumes static corpora and relies on hand-tuned hyperparameters for granularity (e.g., cluster size, unit length, and graph pruning). Future research is needed on self-supervised or task-aware structure induction that optimizes these choices for downstream multi-hop objectives, as well as incremental maintenance algorithms that keep trees and graphs up to date as documents evolve.

Challenge 3: Generalizable control policies and decision criteria. Current learned or search-

based control policies and verifier/PRM-gated decision rules are often trained on specific datasets and retrievers, and may not generalize across hop distributions, index structures, or domains (Zhang et al., 2024; Ji et al., 2025; Yao et al., 2023). This brittleness limits the practical promise of adaptive retrieval–reasoning beyond benchmark-specific tuning. A promising direction is to cast retrieval–reasoning as a process-level decision problem and train controllers with objectives that jointly reward evidence quality, trajectory coherence, and cost, e.g., via preference learning over trajectories or hybrid RL.

Challenge 4: Robust stop/continue criteria under budgets and shift. Most systems still rely on static hop, token, or latency budgets, while adaptive stopping based on confidence, sufficiency, or value estimates is evaluated on limited benchmarks and remains poorly calibrated under distribution shift (Xu et al., 2025; Park et al., 2025). This matters because over- and under-search simultaneously degrade reliability and efficiency in deployment. An important open problem is to develop stop/continue criteria that remain calibrated across retrievers, corpora, and LLM backbones, and to evaluate them under controlled variations of hop depth and retrieval noise.

7 Conclusion

We have surveyed retrieval–reasoning in multi-hop question answering through the lens of four research questions. We argued that multi-hop QA requires an explicit retrieval–reasoning process (RQ1), introduced a four-axis design space for describing this process (RQ2), and analyzed how axis-level design choices are reported to affect effectiveness, efficiency, and faithfulness on standard benchmarks (RQ3). Finally, we highlighted several open problems in designing and evaluating retrieval–reasoning systems (RQ4). We hope that our framework helps researchers organize existing methods and design new retrieval–reasoning agents in a more principled way.

Limitations

First, while we strive to cover a broad range of multi-hop QA systems, the field is evolving rapidly and our selection is necessarily incomplete. Second, we focus primarily on English, text-based benchmarks and do not systematically cover multilingual or multi-modal settings. Third, our empiri-

cal synthesis relies on numbers reported in original papers, which may be influenced by differences in hyperparameters, compute budgets, or evaluation scripts. Fourth, our axis assignments are the product of manual coding; although a subset of papers was double-coded, residual subjectivity remains, particularly for papers that under-specify their control and stopping logic. Despite these limitations, we believe the four-axis framework provides a useful starting point for analyzing retrieval–reasoning systems.

Acknowledgments

We thank the anonymous reviewers for their constructive feedback.

Use of AI Assistants

AI assistants were used for surface-level writing support (grammar and phrasing). All literature coding, analysis, and claims were produced and verified by the authors.

References

- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, and 1 others. 2024. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 17682–17690.
- Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. 2017. Reading wikipedia to answer open-domain questions. *arXiv preprint arXiv:1704.00051*.
- Payel Das, Ching-Yun Ko, Sihui Dai, Georgios Kollias, Subhajit Chaudhury, and Aurelie Lozano. 2025. Can memory-augmented language models generalize on reasoning-in-a-haystack tasks? *arXiv preprint arXiv:2503.07903*.
- Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2024. [Chain-of-verification reduces hallucination in large language models](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3563–3578, Bangkok, Thailand. Association for Computational Linguistics.
- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitan, Robert Osazuwa Ness, and Jonathan Larson. 2024. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.
- Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Zhao, Ni Lao, Hongrae Lee, Da-Cheng Juan, and 1 others. 2023a. Rarr: Researching and revising what language models say, using language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16477–16508.
- Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. 2023b. Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1762–1777.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yixin Dai, Jiawei Sun, Haofen Wang, and Haofen Wang. 2023c. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2(1).
- Mandy Guo, Joshua Ainslie, David C Uthus, Santiago Ontanon, Jianmo Ni, Yun-Hsuan Sung, and Yinfei Yang. 2022. Longt5: Efficient text-to-text transformer for long sequences. In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 724–736.
- Haoyu Han, Yu Wang, Harry Shomer, Kai Guo, Jiayuan Ding, Yongjia Lei, Mahantesh Halappanavar, Ryan A Rossi, Subhabrata Mukherjee, Xianfeng Tang, and 1 others. 2024. Retrieval-augmented generation with graphs (graphrag). *arXiv preprint arXiv:2501.00309*.
- Jinyi Han, Ying Huang, Ying Liao, Zishang Jiang, Xikun Lu, Haiquan Zhao, Xinyi Wang, Guanghao Zhou, Sihang Jiang, Jiaqing Liang, and 1 others. 2025. Your models have thought enough: Training large reasoning models to stop overthinking. *arXiv preprint arXiv:2509.23392*.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Hong, Zhen Wang, Daisy Wang, and Zhiting Hu. 2023. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 8154–8173.
- Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, and Dong Yu. 2024. [WebVoyager: Building an end-to-end web agent with large multimodal models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6864–6890, Bangkok, Thailand. Association for Computational Linguistics.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. *arXiv preprint arXiv:2011.01060*.
- Zhiyuan Hu, Chumin Liu, Xidong Feng, Yilun Zhao, See-Kiong Ng, Anh Tuan Luu, Junxian He, Pang Wei W Koh, and Bryan Hooi. 2024. Uncertainty of thoughts: Uncertainty-aware planning enhances

- information seeking in llms. *Advances in Neural Information Processing Systems*, 37:24181–24215.
- Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th conference of the european chapter of the association for computational linguistics: main volume*, pages 874–880.
- Binquan Ji, Haibo Luo, Yifei Lu, Yifei Lu, Lei Hei, Jiaqi Wang, Tingjing Liao, Wang Lingyu, Shichao Wang, and Feiliang Ren. 2025. [Resource-friendly dynamic enhancement chain for multi-hop question answering](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 16461–16479, Vienna, Austria. Association for Computational Linguistics.
- Jinhao Jiang, Kun Zhou, Wayne Xin Zhao, and Ji-Rong Wen. 2022. Unikgqa: Unified retrieval and reasoning for solving multi-hop question answering over knowledge graph. *arXiv preprint arXiv:2212.00959*.
- Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7969–7992.
- Ziyan Jiang, Xueguang Ma, and Wenhua Chen. 2024. Longrag: Enhancing retrieval-augmented generation with long-context llms. *arXiv preprint arXiv:2406.15319*.
- Jikun Kang, Xin Zhe Li, Xi Chen, Amirreza Kazemi, Qianyi Sun, Boxing Chen, Dong Li, Xu He, Quan He, Feng Wen, and 1 others. 2024. Mindstar: Enhancing math reasoning in pre-trained llms at inference time. *arXiv preprint arXiv:2405.16265*.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick SH Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *EMNLP (1)*, pages 6769–6781.
- Tushar Khot, Harsh Trivedi, Matthew Finlayson, Yao Fu, Kyle Richardson, Peter Clark, and Ashish Sabharwal. 2022. Decomposed prompting: A modular approach for solving complex tasks. *arXiv preprint arXiv:2210.02406*.
- Minsoo Kim, Victor Bursztyjn, Eunye Koh, Shunan Guo, and Seung-won Hwang. 2024. [RaDA: Retrieval-augmented web agent planning with LLMs](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 13511–13525, Bangkok, Thailand. Association for Computational Linguistics.
- Jinhyuk Lee, Anthony Chen, Zhuyun Dai, Dheeru Dua, Devendra Singh Sachan, Michael Boratko, Yi Luan, Séb Arnold, Vincent Perot, Siddharth Dalmia, Hexiang Hu, Xudong Lin, Panupong Pasupat, Aida Amini, Jeremy R. Cole, Sebastian Riedel, Iftekhar Naim, Ming-Wei Chang, and Kelvin Guu. 2025. [LOFT: Scalable and more realistic long-context evaluation](#). In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 6698–6723, Albuquerque, New Mexico. Association for Computational Linguistics.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, and 1 others. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474.
- Moxin Li, Yong Zhao, Wenxuan Zhang, Shuaiyi Li, Wenya Xie, See Kiong Ng, Tat-Seng Chua, and Yang Deng. 2025a. Knowledge boundary of large language models: A survey. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5131–5157.
- Ruosen Li, Zimu Wang, Son Tran, Lei Xia, and Xinya Du. 2024a. Meqa: A benchmark for multi-hop event-centric question answering with explanations. *Advances in Neural Information Processing Systems*, 37:126835–126862.
- Shaobo Li, Xiaoguang Li, Lifeng Shang, Xin Jiang, Qun Liu, Chengjie Sun, Zhenzhou Ji, and Bingquan Liu. 2021. Hopretriever: Retrieve hops over wikipedia to answer complex questions. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 13279–13287.
- Xinze Li, Yixin Cao, Yubo Ma, and Aixin Sun. 2024b. Long context vs. rag for llms: An evaluation and revisits. *arXiv preprint arXiv:2501.01880*.
- Zhong-Zhi Li, Duzhen Zhang, Ming-Liang Zhang, Jiaxin Zhang, Zengyan Liu, Yuxuan Yao, Haotian Xu, Junhao Zheng, Pei-Jie Wang, Xiuyi Chen, and 1 others. 2025b. From system 1 to system 2: A survey of reasoning large language models. *arXiv preprint arXiv:2502.17419*.
- Yaobo Liang, Chenfei Wu, Ting Song, Wenshan Wu, Yan Xia, Yu Liu, Yang Ou, Shuai Lu, Lei Ji, Shaoguang Mao, and 1 others. 2024. Taskmatrix. ai: Completing tasks by connecting foundation models with millions of apis. *Intelligent Computing*, 3:0063.
- Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let’s verify step by step. In *The Twelfth International Conference on Learning Representations*.
- Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pages 9004–9017.

- Vaibhav Mavi, Anubhav Jangra, Adam Jatowt, and 1 others. 2024. Multi-hop question answering. *Foundations and Trends® in Information Retrieval*, 17(5):457–586.
- Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, and 1 others. 2021. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *bmj*, 372.
- Jaewan Park, Solbee Cho, and Jay-Yoon Lee. 2025. Stop-rag: Value-based retrieval control for iterative rag. *arXiv preprint arXiv:2510.14337*.
- Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah Smith, and Mike Lewis. 2023. [Measuring and narrowing the compositionality gap in language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711, Singapore. Association for Computational Linguistics.
- Peng Qi, Xiaowen Lin, Leo Mehr, Zijian Wang, and Christopher D. Manning. 2019. [Answering complex open-domain questions through iterative query generation](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2590–2602, Hong Kong, China. Association for Computational Linguistics.
- Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D Manning. 2024. Raptor: Recursive abstractive processing for tree-organized retrieval. In *The Twelfth International Conference on Learning Representations*.
- Yongliang Shen, Kaitao Song, Xu Tan, Wenqi Zhang, Kan Ren, Siyu Yuan, Weiming Lu, Dongsheng Li, and Yueteng Zhuang. 2024. Taskbench: Benchmarking large language models for task automation. *Advances in Neural Information Processing Systems*, 37:4540–4574.
- Zhengliang Shi, Shuo Zhang, Weiwei Sun, Shen Gao, Pengjie Ren, Zhumin Chen, and Zhaochun Ren. 2024. Generate-then-ground in retrieval-augmented generation for multi-hop question answering. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7339–7353.
- Weihang Su, Yichen Tang, Qingyao Ai, Zhijing Wu, and Yiqun Liu. 2024. [DRAGIN: Dynamic retrieval augmented generation based on the real-time information needs of large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12991–13013, Bangkok, Thailand. Association for Computational Linguistics.
- Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Na Zou, and 1 others. 2025. Stop overthinking: A survey on efficient reasoning for large language models. *arXiv preprint arXiv:2503.16419*.
- Haitian Sun, Tania Bedrax-Weiss, and William Cohen. 2019. Pullnet: Open domain question answering with iterative retrieval on knowledge bases and text. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 2380–2390.
- Haitian Sun, Bhuwan Dhingra, Manzil Zaheer, Kathryn Mazaitis, Ruslan Salakhutdinov, and William Cohen. 2018. Open domain question answering using early fusion of knowledge bases and text. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 4231–4242, Brussels, Belgium. Association for Computational Linguistics.
- Zhongxiang Sun, Qipeng Wang, Weijie Yu, Xiaoxue Zang, Kai Zheng, Jun Xu, Xiao Zhang, Yang Song, and Han Li. 2025. Rearter: Retrieval-augmented reasoning with trustworthy process rewarding. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1251–1261.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. Musique: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*, pages 10014–10037.
- Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. 2023. [Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2609–2634, Toronto, Canada. Association for Computational Linguistics.
- Nan Wang, Yongqi Fan, ZongYu Wang, Xuezhi Cao, Xinyan He, Haiyun Jiang, Tong Ruan, Jingping Liu, and 1 others. 2025a. Kg-o1: Enhancing multi-hop question answering in large language models via knowledge graph integration. *arXiv preprint arXiv:2508.15790*.
- Xinyu Wang, Yanzheng Xiang, Lin Gui, and Yulan He. 2025b. Pecan: Llm-guided dynamic progress control with attention-guided hierarchical weighted graph for long-document qa. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 13317–13335.

- Tomer Wolfson, Mor Geva, Ankit Gupta, Matt Gardner, Yoav Goldberg, Daniel Deutch, and Jonathan Berant. 2020. [Break it down: A question understanding benchmark](#). *Transactions of the Association for Computational Linguistics*, 8:183–198.
- Yu Xia, Subhojyoti Mukherjee, Zhouhang Xie, Junda Wu, Xintong Li, Ryan Aponte, Hanjia Lyu, Joe Barrow, Hongjie Chen, Franck Dernoncourt, and 1 others. 2025. From selection to generation: A survey of llm-based active learning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14552–14569.
- Roy Xie, Junlin Wang, Paul Rosu, Chunyuan Deng, Bolun Sun, Zihao Lin, and Bhuwan Dhingra. 2025. Knowing when to stop: Dynamic context cut-off for large language models. *arXiv preprint arXiv:2502.01025*.
- Yuxi Xie, Anirudh Goyal, Wenyue Zheng, Min-Yen Kan, Timothy P Lillicrap, Kenji Kawaguchi, and Michael Shieh. 2024. Monte carlo tree search boosts reasoning via iterative preference learning. *arXiv preprint arXiv:2405.00451*.
- Wenhan Xiong, Xiang Lorraine Li, Srini Iyer, Jingfei Du, Patrick Lewis, William Yang Wang, Yashar Mehdad, Wen-tau Yih, Sebastian Riedel, Douwe Kiela, and 1 others. 2020. Answering complex open-domain questions with multi-hop dense retrieval. *arXiv preprint arXiv:2009.12756*.
- Ran Xu, Hui Liu, Sreyashi Nag, Zhenwei Dai, Yaochen Xie, Xianfeng Tang, Chen Luo, Yang Li, Joyce C. Ho, Carl Yang, and Qi He. 2025. [SimRAG: Self-improving retrieval-augmented generation for adapting large language models to specialized domains](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11534–11550, Albuquerque, New Mexico. Association for Computational Linguistics.
- Diji Yang, Jinmeng Rao, Kezhen Chen, Xiaoyuan Guo, Yawen Zhang, Jie Yang, and Yi Zhang. 2024. Im-rag: Multi-round retrieval-augmented generation through learning inner monologues. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 730–740.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*, pages 2369–2380.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. 2022. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*.
- Donghan Yu, Chenguang Zhu, Yuwei Fang, Wenhao Yu, Shuohang Wang, Yichong Xu, Xiang Ren, Yiming Yang, and Michael Zeng. 2022. Kg-fid: Infusing knowledge graph in fusion-in-decoder for open-domain question answering. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)*, pages 4961–4974.
- Jiahao Zhang, Haiyang Zhang, Dongmei Zhang, Liu Yong, and Shen Huang. 2024. End-to-end beam retrieval for multi-hop question answering. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1718–1731.
- Ningning Zhang, Chi Zhang, Zhizhong Tan, Xingxing Yang, Weiping Deng, and Wenyong Wang. 2025. Credible plan-driven rag method for multi-hop question answering. *arXiv preprint arXiv:2504.16787*.
- Qingfei Zhao, Ruobing Wang, Yukuo Cen, Daren Zha, Shicheng Tan, Yuxiao Dong, and Jie Tang. 2024. Longrag: A dual-perspective retrieval-augmented generation paradigm for long-context question answering. *arXiv preprint arXiv:2410.18050*.
- Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc Le, and 1 others. 2022. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625*.
- Andrew Zhu, Alyssa Hwang, Liam Dugan, and Chris Callison-Burch. 2024. Fanoutqa: Multi-hop, multi-document question answering for large language models. *arXiv preprint arXiv:2402.14116*.

A PRISMA-style Literature Collection

To identify existing surveys and overviews related to multi-hop question answering (QA), we followed a PRISMA-style (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) procedure adapted to the NLP setting (Page et al., 2021).

Scope and time window. Our goal was to collect survey- or review-type papers that discuss multi-hop QA, multi-step or compositional QA, or widely used multi-hop benchmarks (e.g., HotpotQA, 2WikiMultiHopQA, MuSiQue). We restricted our search to publications from 2022 on-

wards, so as to focus on work in the LLM era and recent retrieval-augmented systems.

Data sources. We queried multiple scholarly search engines and digital libraries, including Google Scholar (using `allintitle` and keyword queries), the ACL Anthology, arXiv, and conference proceedings accessible through ACM and IEEE portals. When a candidate paper appeared in multiple sources, we treated it as a single record.

Search queries. For each source, we issued the following keyword combinations (illustrated here in a Google Scholar-style syntax), always with the time filter set to *since 2022*:

- `allintitle: (survey OR review OR taxonomy OR "systematic review" OR tutorial) ("multi-hop question answering" OR multihop OR "multi hop")`
- `("multi-hop question answering" OR multihop OR "multi hop" OR "multi-step reasoning" OR "complex question answering" OR "compositional question answering") (survey OR review OR taxonomy OR "systematic review" OR tutorial) (NLP OR "natural language processing")`
- `("HotpotQA" OR "2WikiMultiHopQA" OR "MuSiQue" OR "IIRC" OR "ComplexWebQuestions" OR "QAngaroo" OR "MedHopQA") (survey OR review OR taxonomy OR tutorial OR overview)`

These queries were chosen to (1) capture surveys explicitly labelled as such in the title, (2) retrieve broader surveys and tutorials that cover multi-hop or complex QA within NLP, and (3) find overview papers centered around standard multi-hop benchmarks.

Screening and eligibility. We first merged results from all queries and removed exact duplicates based on title and venue. We then screened titles and abstracts to exclude papers that were not surveys, reviews, taxonomies, tutorials, or overviews (e.g., individual method papers, purely theoretical work, or non-NLP applications). In a second pass, we read introductions and methodology sections of the remaining papers to check that multi-hop or complex QA was a primary focus rather than a minor side topic. We retained only

English-language papers that discussed multi-hop QA, multi-step/compositional QA, or the above benchmarks in sufficient detail to be useful for positioning our work.

Data extraction and use in this survey. For each included paper we recorded meta-data such as publication year, venue, target tasks (multi-hop QA, reasoning LLMs, RAG, benchmarks), and the main organisational axes used by the authors (e.g., model architecture, prompting strategy, training regime). These annotations were used in Section 2 to (1) situate our four-component design framework relative to existing taxonomies and (2) ensure that our discussion of multi-hop QA and reasoning surveys is comprehensive with respect to the 2022–2025 literature. Beyond this PRISMA-style survey collection, individual method papers (e.g., specific retrieval–reasoning systems or search algorithms) were added through targeted searches, forward and backward citation tracing, and domain expertise.

B Paper Coding and Topical Distribution

For the analysis in Section 3 we coded each retrieved multi-hop QA / RAG paper along five topical dimensions:

- **T1: Definitions and boundaries of “multi-hop”.** Papers that explicitly discuss what counts as multi-hop reasoning, how many supporting facts or hops are required, or how to distinguish genuine multi-hop questions from artifacts.
- **T2: Decomposition strategies.** Papers that introduce explicit intermediate representations such as chain-of-thought, question decomposition, sub-goal planning, programs/graphs/templates, or MCTS-style search.
- **T3: Evidence sufficiency and faithfulness.** Papers that focus on guaranteeing sufficient support for answers and on aligning reasoning chains with retrieved evidence, often via supporting-fact metrics, verification modules, or LLM-as-judge diagnostics.
- **T4: Retrieval–reasoning coupling.** Papers whose main contribution lies in how retrieval and reasoning are interleaved or controlled, including interleaved CoT–retrieval

loops (Trivedi et al., 2023), beam-based multi-hop retrievers (Zhang et al., 2024), hierarchical or graph-aware retrieval controllers (Sarathi et al., 2024; Edge et al., 2024), and sufficiency-aware multi-round RAG (Xu et al., 2025; Ji et al., 2025).

- **T5: Evaluation and diagnostics.** Papers that propose new diagnostics, process-level metrics, or robustness tests for multi-hop reasoning.

Among all coded papers, topic T4 (retrieval–reasoning coupling) constitutes the largest high-fit group (104 of 252 papers, 41%), followed by T2 (decomposition strategies) and T5 (evaluation and diagnostics); see Table 2. This distribution motivates our decision to take the retrieval–reasoning procedure as the primary object of analysis in the main text.

B.1 Coding Methodology

We constructed a paper corpus through keyword-based search (Google Scholar, ACL Anthology, arXiv, AAAI, ICLR, NeurIPS, ICML, SIGIR, KDD, CVPR, and ICCV) followed by forward/backward citation tracing. After deduplication, we screened titles and abstracts to retain work on multi-hop QA, retrieval-augmented QA, and agentic retrieval–reasoning. For each included paper, we extracted metadata (year, task/benchmarks, supervision, and reported metrics) and coded the method using our four-component schema: overall execution plan, index structure, control policy, and stop/continue criteria. We also assigned each paper to one or more topical dimensions (T1–T5). To improve consistency, a subset of papers was double-coded and disagreements were resolved via discussion; the final codes were then applied to the full set.

From 252 papers to a 104-paper subset. The full corpus contains 252 multi-hop QA/RAG papers used for the broad topical analysis in Section 3. For axis-level distributions and method maps, we focus on a 104-paper subset that provides sufficient methodological detail to instantiate our four components (overall execution plan, index structure, control policy, and stop/continue criteria). This subset is therefore a detail-available, framework-compatible slice of the larger corpus rather than a separate collection.

Topic	High	Med	Low	Total
T1: Definitions / boundaries	6	37	209	252
T2: Decomposition strategies	32	33	187	252
T3: Sufficiency / faithfulness	17	51	184	252
T4: Retrieval–reasoning coupling	104	14	134	252
T5: Evaluation / diagnostics	29	38	185	252

Table 2: Number of papers coded as High / Medium / Low fit for each topical dimension. Retrieval–reasoning coupling (T4) is the largest group in terms of high-fit papers, with 104 of 252 (about 41%) methods focusing primarily on how retrieval and reasoning are coupled.

C Distributions of Design Axes

We summarize corpus-level statistics for the 104-paper subset. Tables 5–8 report single-label distributions (each paper is assigned to exactly one combination), while Table 9 reports the corresponding multi-label marginals.

C.1 Overall Execution Plan

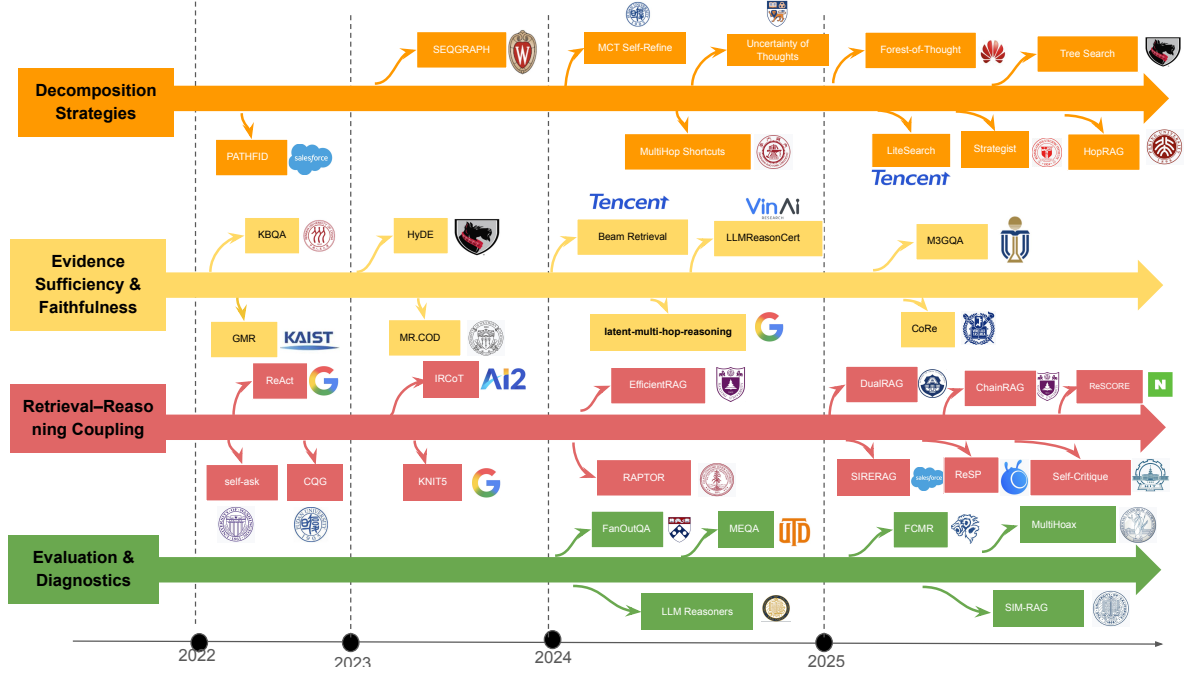


Figure 4: A taxonomy of major research topics in multi-hop question answering and retrieval–reasoning systems from 2022 to 2025. Methods are grouped by decomposition strategies, evidence sufficiency and faithfulness, retrieval–reasoning coupling, and evaluation and diagnostics, illustrating how different lines of work populate these dimensions over time.

Dataset	Type	#papers	Typical role in Topic 4 papers
HotpotQA	Wiki 2-hop RC (supporting facts)	29	Core multi-hop benchmark for retrieval–reasoning loops and supporting-fact faithfulness.
2WikiMultiHopQA	Hybrid KG+Wiki multi-hop QA	15	Tests cross-document and KG–text reasoning; usually paired with HotpotQA and MuSiQue.
WebQuestionsSP	KB QA over Freebase (1–2 hops)	14	Standard KG-QA baseline; many Topic 4 systems evaluate graph or KG controllers on it.
MuSiQue	Compositional multi-hop QA (Wiki)	11	Harder multi-hop benchmark built by composing single-hop questions; stresses connected reasoning.
ComplexWebQuestions (CWQ)	Web+KB complex QA (search+SPARQL)	11	Open-web setting with complex questions; used to test decomposition and web-search control.
MRAMG	Multimodal retrieval (image+text)	6	Extends retrieval–reasoning from text-only QA to multimodal RAG and agent-style systems.
StrategyQA	Boolean QA with implicit multi-hop	5	Benchmarks control policies for implicit, commonsense multi-hop reasoning and decomposition.
MetaQA	Movie-domain multi-hop KGQA (1–3 hops)	4	Tests KG traversal policies and beam / search-based controllers over large KGs.
CronQuestions	Temporal KGQA (time-aware)	4	Evaluates controllers that must handle temporal constraints on KG reasoning paths.
HybridQA	Table+text hybrid multi-hop QA	4	Benchmarks retrieval–reasoning over structured tables plus linked Wikipedia passages.

Table 3: Most commonly used datasets in our Topic 4 (retrieval–reasoning coupling) sample. “#papers” counts how many of the 104 coded papers report results on each dataset. Counts are multi-label: a system may be evaluated on several benchmarks, so the column does not sum to 104.

Dataset	Type	#papers
TriviaQA	Open-domain QA (Wiki + web)	3
Qasper	Long-document QA on NLP papers	3
NarrativeQA	Generative narrative QA (stories)	2
GSM8K	Grade-school math reasoning	2
Natural Questions	Open-domain QA from search logs	2
FB15k-237	Knowledge-graph QA / link prediction	2
ALFWorld	Text-based interactive environment	2
EntailmentBank	Multi-step textual entailment proofs	2
MQuAKE-CF-3k / MQuAKE-T	Counterfactual / transfer multi-hop QA	2
FEVER	Wikipedia fact verification	2
CommonsenseQA	Commonsense multiple-choice QA	2
OpenBookQA	Open-book science QA	2

Table 4: Datasets that appear 2–3 times in the Topic 4 sample.

Scheduling pattern (Axis A)	#papers	Share
Retrieve-then-Read only	49	47.1%
Interleaved only	24	23.1%
Interleaved + Plan-then-Execute	8	7.7%
Retrieve-then-Read + Plan-then-Execute	7	6.7%
Interleaved + Test-time Search Scaling	7	6.7%
Test-time Search Scaling only	5	4.8%
Plan-then-Execute + Test-time Search Scaling	1	1.0%
Plan-then-Execute only	1	1.0%
Other / unspecified	2	1.9%
Total	104	100%

Table 5: Distribution of scheduling patterns (Axis A) across all 104 coded methods in our corpus (topic T4: retrieval–reasoning coupling).

C.2 Index Structure

Index structure (Axis B)	#papers	Share
Graph / KG / community	46	44.2%
Flat / candidate list	29	27.9%
Hierarchical / summary tree	3	2.9%
Long-context	3	2.9%
Web / live retrieval	6	5.8%
Multi-modal / other	5	4.8%
Other / unspecified	12	11.5%
Total	104	100%

Table 6: Distribution of index structures (Axis B) across all 104 methods. Graph- and KG-style indices are the most common among retrieval–reasoning coupling papers, followed by flat passage lists.

C.3 Next-Step Execution Plan

Control pattern (Axis C)	#papers	Share
Policy-based	32	30.8%
Rule-based	31	29.8%
Policy-based + Rule-based	14	13.5%
Rule-based + Verifier/PRM-based	7	6.7%
Policy-based + Search-based	5	4.8%
Search-based + Verifier/PRM-based	4	3.8%
Search-based	2	1.9%
Verifier/PRM-based	2	1.9%
Unspecified	2	1.9%
Policy-based + Verifier/PRM-based	1	1.0%
Policy-based + Uncertainty-gated	1	1.0%
Rule-based + Search-based	1	1.0%
Rule-based + Uncertainty-gated	1	1.0%
Policy-based + Rule-based + Verifier/PRM-based	1	1.0%
Total	104	100%

Table 7: Distribution of next-step execution / control patterns across all 104 methods (Axis C).

C.4 Stop / Continue Criteria

Stop / continue pattern (Axis D)	#papers	Share
Budget-based + Heuristic-done	42	40.4%
Budget-based	40	38.5%
Budget-based + Verifier-threshold	10	9.6%
Budget-based + Memory/Progress	5	4.8%
Heuristic-done	5	4.8%
Unspecified	2	1.9%
Total	104	100%

Table 8: Distribution of stop / continue criteria across all 104 methods (Axis D).

D Marginal Distribution of the Four Components

Table 9 summarizes how the four components in our framework are instantiated across the 104 multi-hop QA / RAG papers in our Topic 4 sample. Along Axis A (overall execution plan), more than half of the systems (56/104, 53.8%) still follow a frozen-pool “retrieve-then-read” pattern, while 39/104 (37.5%) explicitly interleave retrieval and reasoning, 17/104 (16.3%) implement plan-then-execute strategies, and 13/104 (12.5%) scale test-time search (e.g., Tree-of-Thoughts or MCTS) on top of these patterns.

Axis B (index structure) shows that graph- or KG-style indices are the most common single family, followed by flat text indices, whereas hierarchical summary trees and long-context retrieval units remain rare. A further group of papers operates without a fixed static index, instead relying on web search, multimodal retrieval, or purely parametric knowledge.

For Axis C (next-step execution plan), roughly half of the systems employ a learned control policy, and a similar number still use rule-based action selection such as fixed reasoning rhythms or hand-crafted heuristics; only a small minority rely on fully search-based controllers (beam search, MCTS, or Tree-of-Thoughts).

Finally, Axis D (stop / continue criteria) is dominated by resource-constrained stopping: 97/104 systems (93.3%) impose explicit hop, token, or latency budgets, and 47/104 (45.2%) combine these with simple “done when the path is complete” heuristics or beam-exhaustion rules. Verifier- or PRM-based stopping remains less common but growing, and only a handful of systems explicitly use uncertainty- or confidence-based stopping. These distributions support our claim that most recent progress focuses on richer control and planning on top of relatively conservative index struc-

Axis / choice	#papers	Share of 104
(A) Overall execution plan		
Retrieve-then-read	56	53.8%
Interleaved retrieval + reasoning	39	37.5%
Plan-then-execute	17	16.3%
Test-time search scaling (ToT/MCTS/beam)	13	12.5%
(B) Index structure		
Flat text / candidate list	29	27.9%
Graph / KG / community index	46	44.2%
Hierarchical / summary tree	3	2.9%
Long-context units	3	2.9%
Other (web / multimodal / param.)	23	22.1%
(C) Next-step execution plan: control strategy		
Policy-based control	54	51.9%
Rule-based control	55	52.9%
Search-based control (beam / MCTS / ToT)	12	11.5%
Verifier / PRM-gated control	15	14.4%
Uncertainty-gated control	2	1.9%
(D) Stop / continue criteria		
Budget-based stopping (hop / token / latency)	97	93.3%
Heuristic / done-when-path-complete rules	47	45.2%
Progress-based rules (memory / frontier)	5	4.8%
Verifier- / PRM-based stopping	10	9.6%

Table 9: Distribution of retrieval–reasoning design choices across the 104 multi-hop QA / RAG papers in our Topic 4 sample, aligned with the four components in Figure 1. Counts are multi-label within each block: a single system may instantiate multiple choices along an axis, so percentages do not sum to 100%. Values are the marginals of Tables 5–8.

tures and budgeted stopping schemes.

E Reported Performance of Representative Systems

Dataset	System	Year	Reported performance on this dataset
2WikiMulti HopQA	Credible Plan-Driven RAG Method for Multi-hop Question Answering (PAR-RAG)	2025	EM 71.23, Acc 78.25
2WikiMulti HopQA	KG-o1: Enhancing Multi-hop Question Answering in LLMs with Knowledge Graphs	2025	EM 62.40 / F1 74.45 / Prec. 72.56 / Rec. 79.45
HotpotQA	Resource-Friendly Dynamic Enhancement Chain for Multi-hop QA (DEC)	2025	CoverEM 47.19 / F1 50.96 / Acc 49.60; CoverEM 58.52 / F1 62.11 / Acc 60.32
HotpotQA	KG-o1: Enhancing Multi-hop Question Answering in LLMs with Knowledge Graphs	2025	EM 51.00 / F1 67.11 / Prec. 69.66 / Rec. 68.18
MuSiQue	End-to-End Beam Retrieval for Multi-Hop Question Answering	2024	MuSiQue-Ans (test): Answer F1 69.2, Support F1 91.4 (human Sp 93.9)
IIRC	Interleaving Retrieval with Chain-of-Thought Reasoning (IRCoT)	2023	Flan-T5-XXL: IRCoT vs. OneR → +2.5 F1, with gains holding OOD and at smaller model scales
StrategyQA	LLM-based Search Assistant with Holistically Guided MCTS	2025	EM/F1 77.67 / 77.67 (GPT-4o-mini, best setting)
FanOutQA	LLM-based Search Assistant with Holistically Guided MCTS	2025	Acc 58.38; R-1/2/L 55.02 / 35.45 / 49.40